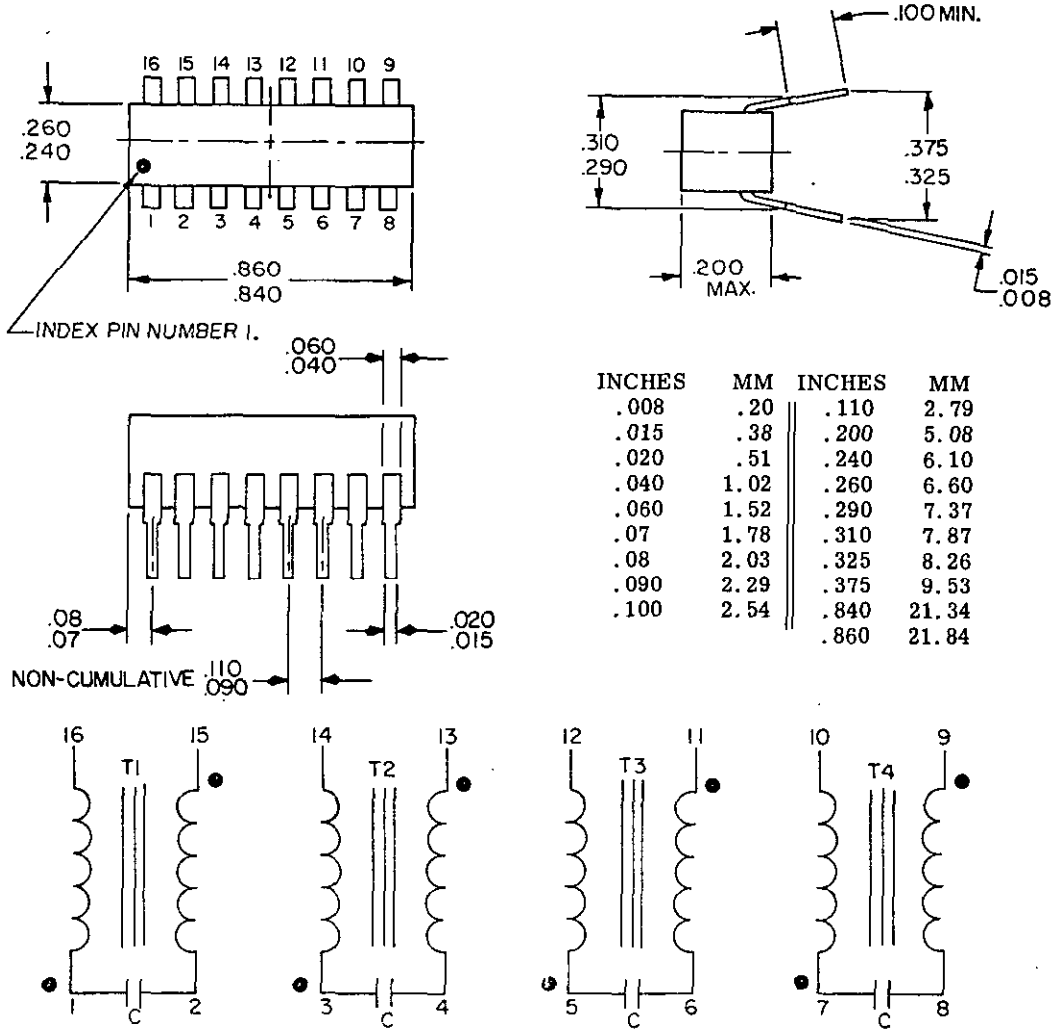
MILITARY SPECIFICATION SHEET

TRANSFORMERS, PULSE, LOW POWER,

TYPE TP7SX1100MZ

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-21038.



INCHES	MM	INCHES	MM
.008	.20	.110	2.79
.015	.38	.200	5.08
.020	.51	.240	6.10
.040	1.02	.260	6.60
.060	1.52	.290	7.37
.07	1.78	.310	7.87
.08	2.03	.325	8.26
.090	2.29	.375	9.53
.100	2.54	.840	21.34
		.860	21.84

Circuit diagram and schematic

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The capacitance (C) is optional.

FIGURE 1. Transformer type TP7SX1100MZ.

FSC 5950

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Voltage ratio: 1:1 ±2 percent.

Case: Encapsulated.

Duty cycle: Continuous.

Terminals: Pin type, solderable.

Terminal material: Type K, MIL-STD-1276.

Weight: 1 ounce maximum.

Terminal strength: Method 2004, MIL-STD-883, test condition B2.

Dielectric withstanding voltage:
At atmospheric pressure: Method 301, MIL-STD-202. 50 volts rms, 60 Hz, for 5 seconds between windings.
Barometric pressure (reduced): Not applicable.

Reliability conditioning (burn-in): Not applicable.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Not less than 1 G ohm at 50 volts dc primary to secondary.

Electrical characteristics: See table I.
Volt-microseconds rating (ET): The ET rating shall be greater than or equal to 1.0 volt-microseconds over the operating temperature range of -55° C to 125° C. The circuit shall be connected as shown in figure 2, using a pulse generator capable of supplying a 1 volt, 1 μs pulse at E_{in} with the transformer not connected. Connect R_G and R_1 so that the source impedance is 10 ohms ±5 percent. Measure pulse amplitude of E_o at 0.2 μs, 0.5 μs, and at 0.8 μs. Then the ET constant is defined by: $ET = 0.1 (e_1 + 4e_5 + e_8)$ volt-microseconds.

Where:
E_o = secondary voltage amplitude in volts at time t
t = time of measurement in microseconds
$e_1 = E_o$ at 0.2 μs
$e_5 = E_o$ at 0.5 μs
$e_8 = E_o$ at 0.8 μs

Test circuit: See figure 2.

Life (expectancy "x"): Applicable.

Temperature range:
Operating: -55° C to 130° C.
Storage: -65° C to 130° C.

Altitude: 70,000.

Vibration high frequency: Method 204, test condition D, MIL-STD-202. Prior to testing, the parts shall be placed in an oven and preheated to +95° C plus one-half the temperature rise. Vibration in each plane must begin within 5 minutes after removal of the units from the oven.

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Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Moisture resistance: Polarizing voltage not applicable.

Part number: M21038/14-(dash number from table I).

Cross reference: See table II.

TABLE I. Electrical characteristics.

Dash no.	Primary inductance (±20%)	Primary ET constant (min)	Risetime (max)	Capacitor C (approx)	Interwinding capacitance (max)	Leakage inductance (max)	Pulse repetition rate (nom)	Pulse amplitude		DCR	
								Min	Max	Pri	Sec
	μH	Vμs	μs	pF	pF	μH	KHz	Volts		ohms	
001	11.0	1.0	0.05	---	30	1.0	10	1	2	0.5	0.5
002	11.0	1.0	0.05	100	30	1.0	10	1	2	0.5	0.5

TABLE II. Cross reference.

Dash no.	Manufacturer's part number	Manufacturer's identification code
001	905548-1 B41W9D 23Z49	06481 90095 05571
002	905548-2 B41X9A 908Z3	06481 90095 05571

Transformer under test.

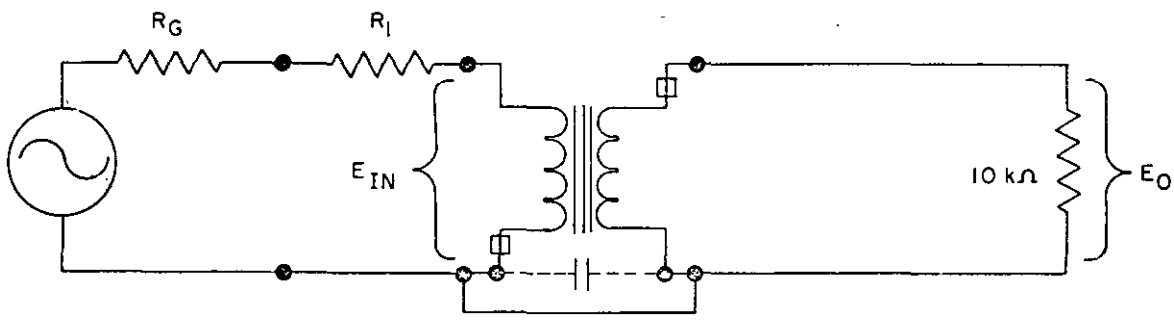


FIGURE 2. Test circuit.

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